Biochemistry Cume February 9, 2001

1. (15 points) Explain two or three molecular properties that are important for lipid function and that cause lipids to be quite different from all of the other major classes of biomolecules.

2. (15 points) If the signal peptide portion of a gene for a normally secreted protein has been deleted from a cell's chromosome but the remainder of the gene is intact, what will most probably be the final destination for the newly synthesized protein? Justify your answer with a short explanation.

3. (45 points) Calculate the electrochemical potential at 37 °C for each case below:

a. intracellular $[Ca^{++}]$ = 15 mM, extracellular $[Ca^{++}]$ = 200 mM, membrane potential = -60 mV (inside negative).

b. intracellular $[K^{+}]$ = 400 mM, extracellular $[K^{+}]$ = 25 mM, membrane potential = -60 mV (inside negative).

4. (10 points) Why can a single α-helix span a lipid bilayer membrane in a stable transmembrane configuration, whereas a single β-strand cannot?

5. (15 points) Explain this observation: A membrane-bound receptor of MW 80,000 daltons is isolated from a cell membrane, solubilized with detergent, purified and reconstituted into lipid vesicles. Following brief treatment of the resulting sample with chymotrypsin, the protein content is analyzed by SDS gel electrophoresis. Equimolar amounts of protein with MW of 80,000, 50,000 and 30,000 are observed. Why?

Possibly useful data: Faraday's constant, 96,485 joule mol^{-1} $volt^{-1}$;
gas constant, 8.315 joule mol^{-1} deg^{-1}.